PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Practice Midterm C - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Choose 3 of the following 4 problems. Complete every part of each chosen problem.

Write all rough calculations and final answers on separate paper. Show your working.

Original source numbering is retained inside each question. Answer the entire cropped question.

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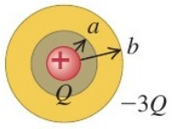
1. [____] Textbook 15e - Question 23.75

23.75 • An alpha particle with kinetic energy 9.50 MeV (when far away) collides head-on with a lead nucleus at rest. What is the distance of closest approach of the two particles? (Assume that the lead nucleus remains stationary and may be treated as a point charge. The atomic number of lead is 82. The alpha particle is a helium nucleus, with atomic number 2.)

2. [____] Textbook 15e - Question 22.42

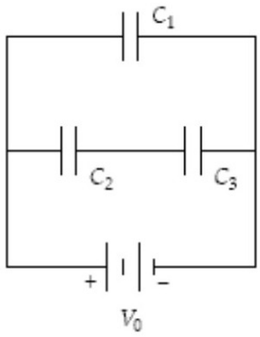
22.42 • A conducting spherical shell with inner radius a and outer radius b has a positive point charge Q located at its center. The total charge on the shell is $-3Q$, and it is insulated from its surroundings (**Fig. P22.42**). (a) Derive expressions for the electric-field magnitude E in terms of the distance r from the center for the regions $r < a$, $a < r < b$, and $r > b$. What is the surface charge density (b) on the inner surface of the conducting shell; (c) on the outer surface of the conducting shell? (d) Sketch the electric field lines and the location of all charges. (e) Graph E as a function of r.

Figure **P22.42**



3. [____] Tutorial 7 - Question 2

2. What is the potential difference across C_2 when $C_1 = 5.0\ \mu$F, $C_2 = 15\ \mu$F, $C_3 = 30\ \mu$F, and $V_0 = 24$ V?



a. 21 V
b. 19 V
c. 16 V
d. 24 V
e. 8.0 V

4. [____] Tutorial 7 - Question 7

7. A parallel plate capacitor of capacitance C_0 has plates of area A with separation d between them. When it is connected to a battery of voltage V_0, it has charge of magnitude Q_0 on its plates. It is then disconnected from the battery and the space between the plates is filled with a material of dielectric constant 3. After the dielectric is added, the magnitudes of the charge on the plates and the potential difference between them are
   a. $\frac{1}{3}Q_0, \frac{1}{3}V_0$.
   b. $Q_0, \frac{1}{3}V_0$.
   c. Q_0, V_0.
   d. $Q_0, 3V_0$.
   e. $3Q_0, 3V_0$.